Tanmay Bhuskute

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SUMMARY

Software Engineer with production experience designing and operating scalable microservices, REST APIs, and cloud infrastructure using Python, SQL and NoSQL databases, and AWS. Built CI/CD pipelines, Kubernetes-based deployment platforms, and monitoring workflows supporting 40+ production environments, with a track record of Agile delivery and production support.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Database Theory, Deep Learning.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Object-Oriented Programming, Data Mining, Machine Learning.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, SQL, Java, Go, C/C++.

Backend & APIs: Node.js, Django, Flask, FastAPI, REST APIs, Microservices Architecture, WebSockets.

Databases: PostgreSQL, MySQL, DynamoDB, MongoDB, Oracle DB, Redis, Elasticsearch.

Cloud & DevOps: AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes, ArgoCD, Helm, Terraform, GitLab CI/CD, Jenkins, SonarQube, Azure, GCP.

Testing & Practices: JUnit, Pytest, Jest, Agile/Scrum, TDD, Design Patterns, Production Support.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against a published VLM baseline across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Designed and operated production microservices on **Kubernetes**, migrating **40+ environments** from process-based deployments using ArgoCD, Helm, GitLab CI/CD, and Amazon EFS, tracking state across **DynamoDB** and relational stores.
- Automated infrastructure workflows with Python (Boto3) and REST APIs, replacing legacy Ruby scripts and cutting deployment time **40%** across AWS environments.
- Executed **44 production upgrades** in 4 months, using **CloudWatch** and **Datadog** for live monitoring, alerting, and diagnosis across Linux, EC2, databases, and services.
- Designed **Ansible-based Blue-Green deployment** framework, cutting production upgrade downtime **50%** with automated rollback and recovery.
- Served as **Quality Coach** on the Agile Release Train, raising automated test coverage **35% to 78%** via SonarQube CI/CD quality gates; mentored **5 engineers** through onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Express, MCP, Docker] May 2026

- Built full-stack TypeScript app (React SPA + Express 5 REST API), designing **17 modular tool APIs** with Zod schema validation at every data boundary, eliminating a class of runtime type errors.
- Implemented PDF-to-structured-data pipeline with Vitest unit and integration tests and axe-core accessibility coverage, supporting production-quality test practices.

NYC Mobility Forecasting [Python, FastAPI, PostgreSQL, Redis, Docker] Jul 2026

- Built RESTful API (FastAPI) serving real-time predictions, integrated with **PostgreSQL** and **Redis** caching, containerized with Docker; cut prediction error (RMSE) **43%** over baseline across 263 zones.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).